

HIV Analysis

STA198 Team: Natalie, Ella, Jessica, Ali

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```
rm(list = ls())
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.0      v readr      2.1.6
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.2      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()    masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(janitor)
```

```
##
```

```
## Attaching package: 'janitor'
```

```
##
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      chisq.test, fisher.test
```

```
deaths <- read_csv("Data/no_of_deaths_by_country_clean.csv")
```

```
## Rows: 510 Columns: 7
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): Country, Count, WHO Region
```

```
## dbl (4): Year, Count_median, Count_min, Count_max
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
hiv <- read_csv("Data/no_of_people_living_with_hiv_by_country_clean.csv")
```

```
## Rows: 680 Columns: 7
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): Country, Count, WHO Region
```

```
## dbl (4): Year, Count_median, Count_min, Count_max
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```

art <- read_csv("Data/art_coverage_by_country_clean.csv")

## Rows: 170 Columns: 11
## -- Column specification -----
## Delimiter: ","
## chr (5): Country, Reported number of people receiving ART, Estimated number ...
## dbl (6): Estimated number of people living with HIV_median, Estimated number...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

deaths_clean <- deaths |>
  clean_names() |>
  group_by(country) |>
  summarise(
    deaths = first(count_median),
    .groups = "drop"
  )

hiv_clean <- hiv |>
  clean_names() |>
  group_by(country) |>
  summarise(
    hiv_population = first(count_median),
    .groups = "drop"
  )

art_clean <- art |>
  clean_names() |>
  select(
    country,
    art_coverage = estimated_art_coverage_among_people_living_with_hiv_percent_median
  )

final_df <- deaths_clean |>
  inner_join(hiv_clean, by = "country") |>
  inner_join(art_clean, by = "country")

write_csv(final_df, "Data/final_df.csv")

glimpse(final_df)

## Rows: 170
## Columns: 4
## $ country      <chr> "Afghanistan", "Albania", "Algeria", "Angola", "Argenti~
## $ deaths       <dbl> 500, NA, 200, 14000, 1700, 200, 200, NA, NA, 200, NA, 5~
## $ hiv_population <dbl> 7200, NA, 16000, 330000, 140000, 3500, 28000, NA, NA, 6~
## $ art_coverage  <dbl> 13, NA, 81, 27, 61, 53, 83, NA, NA, 52, NA, 22, 50, 59,~

```

Description

I created a merged, final dataset to use for analysis with cleaned and standardized columns (contains 170 countries and 4 variables). Each row represents a single country. The dataset was constructed by merging three country-level datasets sourced from WHO-based HIV/AIDS indicators available on Kaggle. All variables are continuous numerical estimates, and represent median values reported in the original datasets. The variables included in the final dataset are: - HIV/AIDS deaths (deaths) - Number of people living with HIV

(hiv_population) - ART coverage among people living with HIV (art_coverage)

```
colSums(is.na(final_df))
```

```
##      country      deaths hiv_population  art_coverage
##           0           37           32           34
```

```
colMeans(is.na(final_df))
```

```
##      country      deaths hiv_population  art_coverage
## 0.0000000 0.2176471 0.1882353 0.2000000
```

Description

The dataset contains missing values across all three main variables: HIV/AIDS deaths (37 missing, ~21.8%), number of people living with HIV (32 missing, ~18.8%), and ART coverage (34 missing, 20%); no missing data for countries. This is most likely due to incomplete reporting in certain countries within the original WHO-based dataset (real-world data is often incomplete). Missing data was not cleaned or removed, just be wary of it and decide what to do with it in the next stages (if anything).

```
final_df |>
  arrange(desc(deaths)) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##   country      deaths hiv_population art_coverage
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 South Africa  71000      7700000      62
## 2 Mozambique   54000      2200000      56
## 3 Nigeria      53000      1900000      53
## 4 Indonesia    38000       640000      17
## 5 Kenya      25000      1600000      68
## 6 United Republic of Tanzania 24000      1600000      71
## 7 Uganda       23000      1400000      72
## 8 Zimbabwe     22000      1300000      88
## 9 Cameroon     18000       540000      52
## 10 Thailand     18000       480000      75
```

Description

Shows top 10 countries with the most HIV/AIDS deaths.

```
final_df |>
  filter(!is.na(deaths)) |>
  arrange(deaths) |>
  slice(1:10)
```

```
## # A tibble: 10 x 4
##   country      deaths hiv_population art_coverage
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 Barbados     100        3000       50
## 2 Bhutan        100       1300       37
## 3 Bosnia and Herzegovina 100        500       67
## 4 Bulgaria      100       3500       41
## 5 Cabo Verde    100       2400       89
## 6 Comoros        100        200       79
## 7 Croatia        100       1600       75
```

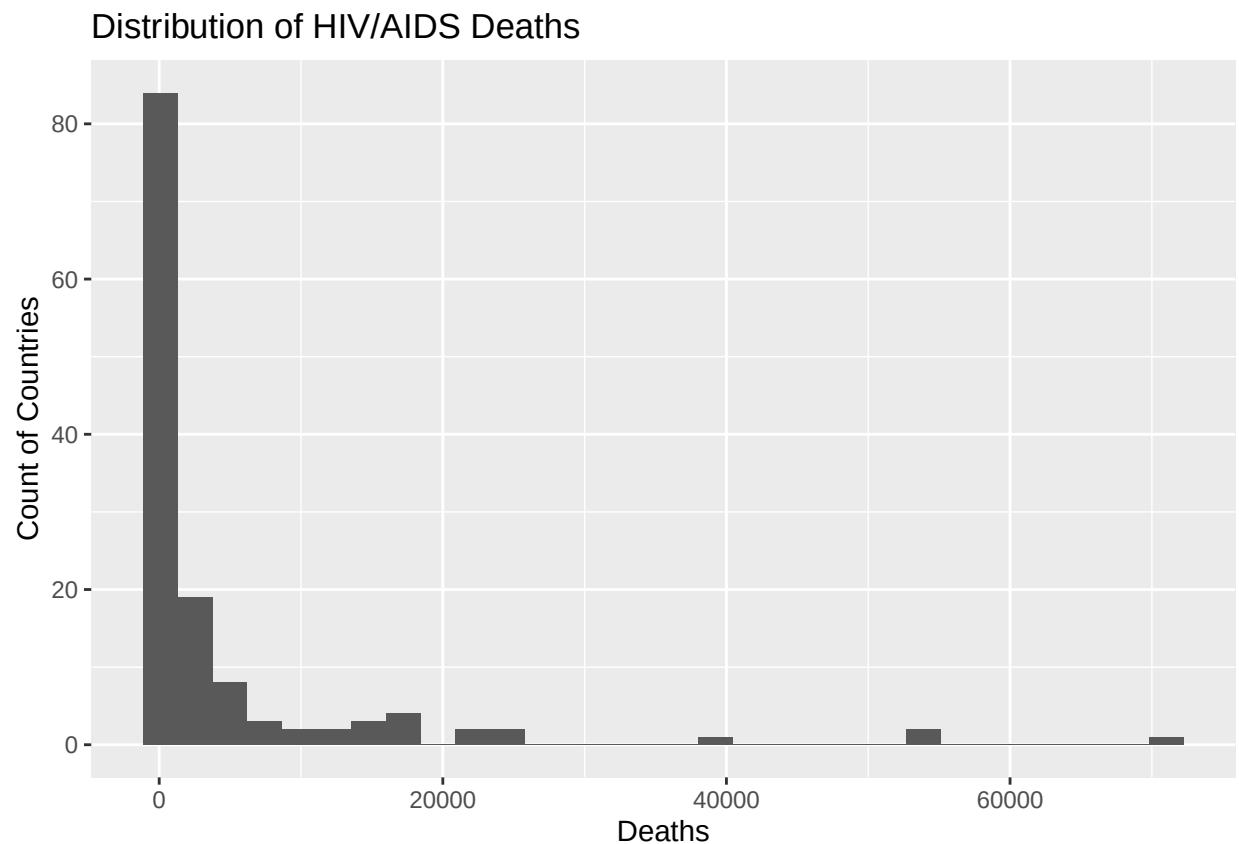
##	8	Czechia	100	4400	60
##	9	Denmark	100	6200	89
##	10	Estonia	100	7400	59

Description

Shows top 10 countries with the least HIV/AIDS deaths.

```
ggplot(final_df, aes(x = deaths)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of HIV/AIDS Deaths", x = "Deaths", y = "Count of Countries")
```

```
## Warning: Removed 37 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

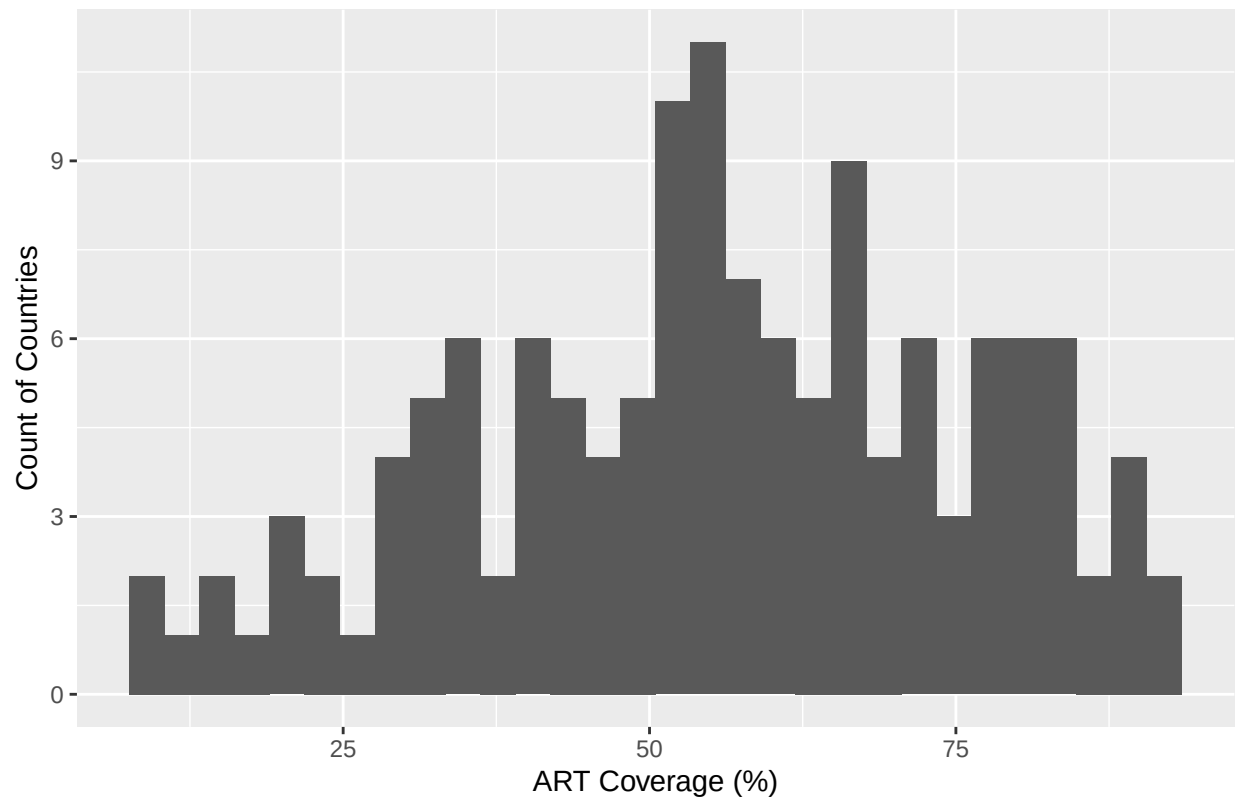


```
## Description Shows the distribution of deaths across countries.
```

```
ggplot(final_df, aes(x = art_coverage)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of ART Coverage", x = "ART Coverage (%)", y = "Count of Countries")
```

```
## Warning: Removed 34 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

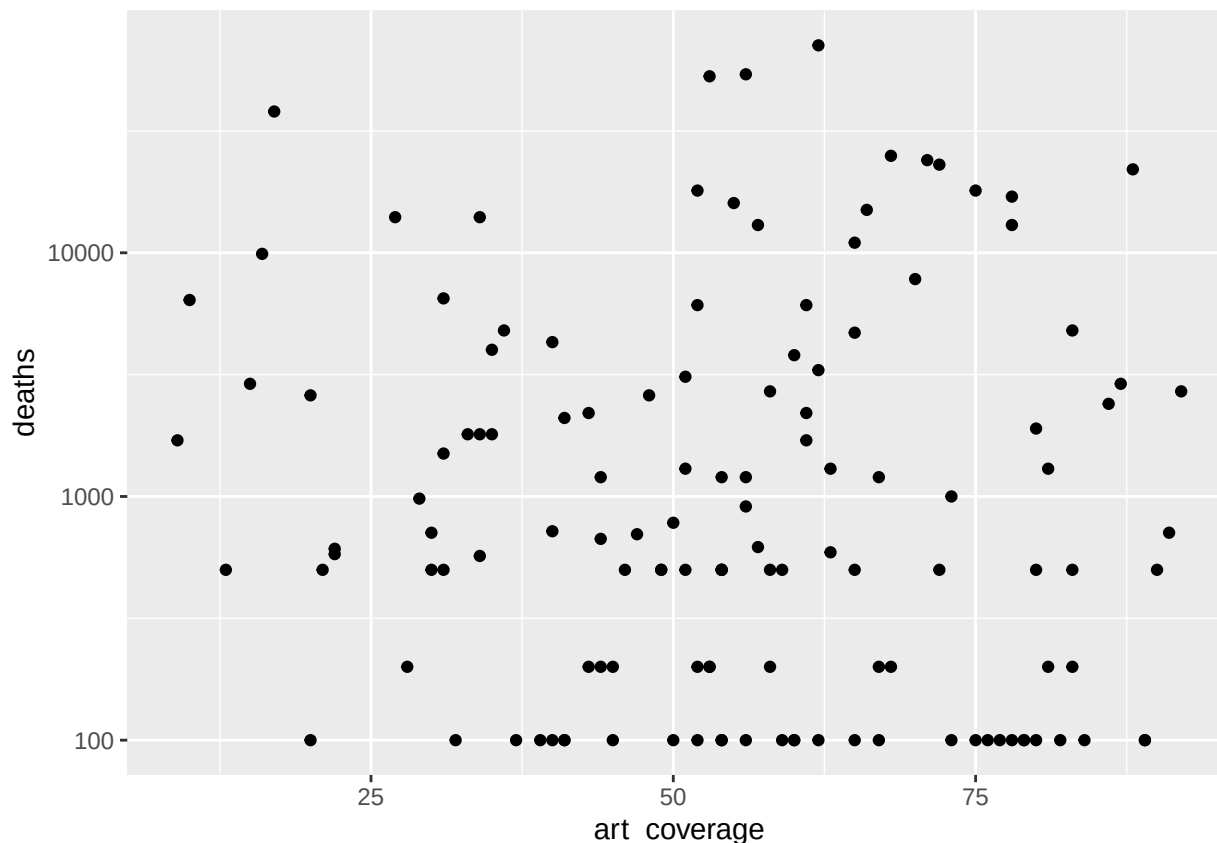
Distribution of ART Coverage



Description Shows the distribution of treatment access (ART coverage %) across countries.

```
ggplot(final_df, aes(x = art_coverage, y = deaths)) +
  geom_point() +
  scale_y_log10()
```

Warning: Removed 39 rows containing missing values or values outside the scale range
(`geom\_point()`).



Description Shows the relationship between ART coverage (%) and HIV/AIDS deaths across countries, where each point represents a country. Note that I used log scaling for the y-axis to account for scaling issues and compression of the graph.

Model Building

To examine the relationship between ART coverage and HIV/AIDS deaths, I conducted a correlation analysis and multiple linear regression models. I first tested the direct association between ART coverage and deaths, then included the number of people living with HIV as a control variable.

Correlational Test

```
cor(final_df$art_coverage, final_df$deaths, use = "complete.obs")
```

```
## [1] 0.02297943
```

Statistical Analysis:

The correlation between ART coverage and HIV/AIDS was very weak ($r = 0.023$), indicating little to no linear relationship when examined alone. However, this result may be misleading because it does not account for differences in the number of people living with HIV across countries, which can strongly influence death counts. This means that a controlled regression analysis should be conducted.

Simple regression model

```
model1 <- lm(deaths ~ art_coverage, data = final_df)
summary(model1)
```

```
##
## Call:
## lm(formula = deaths ~ art_coverage, data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4934  -4400  -3800  -1538   66292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3960.15    2697.69   1.468   0.145
## art_coverage    12.07     46.22   0.261   0.794
##
## Residual standard error: 10580 on 129 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.0005281, Adjusted R-squared:  -0.00722
## F-statistic: 0.06815 on 1 and 129 DF,  p-value: 0.7945
```

Statistical Analysis:

A simple linear regression was conducted to examine the relationship between ART coverage and HIV/AIDS deaths. The results showed no significant relationship between ART coverage and deaths ($\beta = 12.07$, $p = 0.794$). Additionally, the model explained virtually none of the variation in deaths ($R^2 = 0.0005$), indicating that ART coverage alone is not a meaningful predictor of HIV/AIDS mortality. This suggests that other factors, such as the size of the HIV-positive population, may play a more important role.

Controlled regression model

```
model2 <- lm(deaths ~ art_coverage + hiv_population, data = final_df)
summary(model2)

##
## Call:
## lm(formula = deaths ~ art_coverage + hiv_population, data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23510.6  -1746.6  -1210.4   -276.6   28179.1
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.021e+03  1.347e+03   2.985   0.0034 **
## art_coverage  -4.055e+01  2.323e+01  -1.745   0.0833 .
## hiv_population  1.208e-02  6.121e-04  19.732  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5285 on 128 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.7527, Adjusted R-squared:  0.7489
## F-statistic: 194.8 on 2 and 128 DF,  p-value: < 2.2e-16
```

Statistical Analysis:

A multiple linear regression model was used to examine the relationship between ART coverage and HIV/AIDS deaths while controlling for the number of people living with HIV. The results showed that ART coverage had a negative association with deaths ($\beta = -40.55$, $p = 0.083$), suggesting that higher ART coverage is associated with fewer HIV/AIDS deaths. Although this relationship was not statistically significant at the 0.05 level, it indicates a meaningful downward trend.

In contrast, the number of people living with HIV was a highly significant predictor of deaths ($\beta = 0.01208$, $p = 2 \times 10^{-16}$), indicating that countries with larger HIV-positive populations experience substantially more deaths.

The model overall demonstrated strong explanatory power ($R^2 = 0.753$), meaning that approximately 75% of the variation in HIV/AIDS deaths across countries is explained by these variables.

All in all, while ART coverage alone did not appear to influence deaths in the simple model, the controlled regression suggests that increased ART coverage is associated with reduced HIV/AIDS mortality when accounting for differences in HIV pop. size. This supports the hypothesis that expanding access to antiretroviral therapy may help reduce HIV-related deaths globally.

Log model

```
model3 <- lm(log(deaths) ~ art_coverage + log(hiv_population), data = final_df)
summary(model3)
```

```
##
## Call:
## lm(formula = log(deaths) ~ art_coverage + log(hiv_population),
##     data = final_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.61043 -0.38433  0.01353  0.34033  2.16313
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.662964   0.275172  -2.409   0.0174 *
## art_coverage   -0.014334   0.002538  -5.649 9.96e-08 ***
## log(hiv_population) 0.799767   0.023634  33.840 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5792 on 128 degrees of freedom
## (39 observations deleted due to missingness)
## Multiple R-squared:  0.9001, Adjusted R-squared:  0.8986
## F-statistic: 576.8 on 2 and 128 DF,  p-value: < 2.2e-16
```

Statistical Analysis:

A log-transformed multiple regression model was used to better account for skewness in the data and examine the relationship between ART coverage and HIV/AIDS deaths. The results showed... (Ella)

k